

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

print("Libraries Imported Successfully")
```

Libraries Imported Successfully

```
image = cv2.imread("Babe.jpg")

if image is None:
    print("Image not found!")
else:
    print("Image Loaded Successfully")

image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(6,6))
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```

Image Loaded Successfully

Original Image



```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

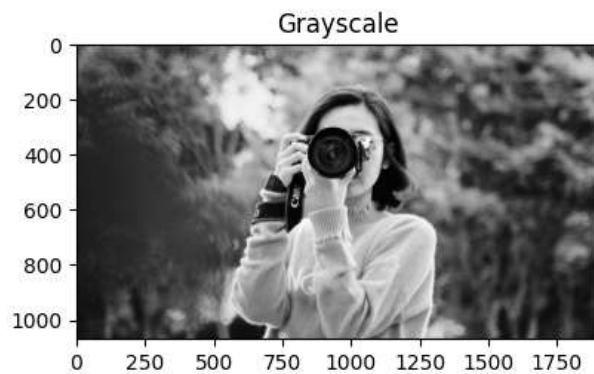
blur = cv2.GaussianBlur(gray, (5,5), 0)

plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(gray, cmap='gray')
plt.title("Grayscale")

plt.subplot(1,2,2)
plt.imshow(blur, cmap='gray')
plt.title("Gaussian Blur")

plt.show()
```



```
_, global_thresh = cv2.threshold(  
    blur,  
    127,  
    255,  
    cv2.THRESH_BINARY  
)
```

```
plt.figure(figsize=(6,6))  
plt.imshow(global_thresh, cmap='gray')  
plt.title("Global Thresholding")  
plt.axis("off")  
plt.show()
```



```
otsu_value, otsu_thresh = cv2.threshold(  
    blur,  
    0,  
    255,  
    cv2.THRESH_BINARY + cv2.THRESH_OTSU  
)
```

```
print("Otsu Threshold Value =", otsu_value)
```

```
plt.figure(figsize=(6,6))  
plt.imshow(otsu_thresh, cmap='gray')  
plt.title("Otsu Thresholding")  
plt.axis("off")  
plt.show()
```

Otsu Threshold Value = 117.0

Otsu Thresholding

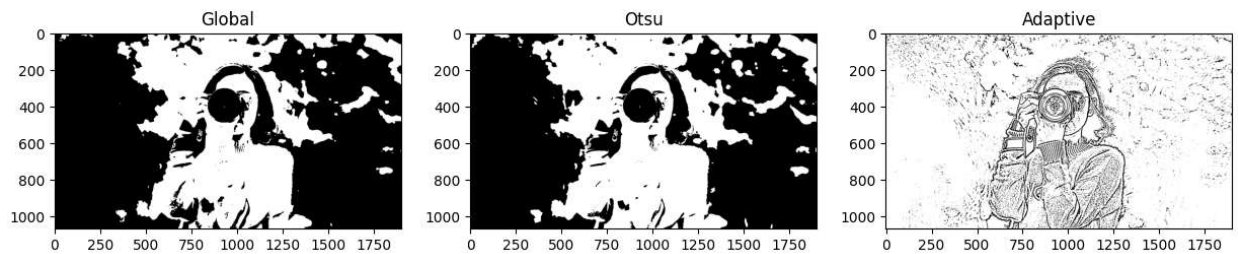


```
adaptive_thresh = cv2.adaptiveThreshold(  
    blur,  
    255,  
    cv2.ADAPTIVE_THRESH_GAUSSIAN_C,  
    cv2.THRESH_BINARY,  
    11,  
    2  
)  
  
plt.figure(figsize=(6,6))  
plt.imshow(adaptive_thresh, cmap='gray')  
plt.title("Adaptive Thresholding")  
plt.axis("off")  
plt.show()
```

Adaptive Thresholding



```
plt.figure(figsize=(15,5))  
  
plt.subplot(1,3,1)  
plt.imshow(global_thresh, cmap='gray')  
plt.title("Global")  
  
plt.subplot(1,3,2)  
plt.imshow(otsu_thresh, cmap='gray')  
plt.title("Otsu")  
  
plt.subplot(1,3,3)  
plt.imshow(adaptive_thresh, cmap='gray')  
plt.title("Adaptive")  
  
plt.show()
```



```

img = image.copy()

gray_ws = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

_, thresh = cv2.threshold(
    gray_ws,
    0,
    255,
    cv2.THRESH_BINARY_INV + cv2.THRESH_OTSU
)

kernel = np.ones((3,3), np.uint8)

opening = cv2.morphologyEx(
    thresh,
    cv2.MORPH_OPEN,
    kernel,
    iterations=2
)

sure_bg = cv2.dilate(opening, kernel, iterations=3)

dist_transform = cv2.distanceTransform(
    opening,
    cv2.DIST_L2,
    5
)

_, sure_fg = cv2.threshold(
    dist_transform,
    0.7 * dist_transform.max(),
    255,
    0
)

sure_fg = np.uint8(sure_fg)

unknown = cv2.subtract(sure_bg, sure_fg)

_, markers = cv2.connectedComponents(sure_fg)

markers = markers + 1

markers[unknown == 255] = 0

markers = cv2.watershed(img, markers)

img[markers == -1] = [255, 0, 0]

plt.figure(figsize=(8,8))
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Watershed Segmentation")
plt.axis("off")
plt.show()

```

Watershed Segmentation



```
pixel_values = image_rgb.reshape((-1, 3))

pixel_values = np.float32(pixel_values)

k = 4

kmeans = KMeans(
    n_clusters=k,
    random_state=0,
    n_init=10
)

labels = kmeans.fit_predict(pixel_values)

centers = np.uint8(kmeans.cluster_centers_)

segmented_data = centers[labels.flatten()]

segmented_image = segmented_data.reshape(image_rgb.shape)

plt.figure(figsize=(8,8))
plt.imshow(segmented_image)
plt.title("K-Means Segmentation")
plt.axis("off")
plt.show()
```

K-Means Segmentation



```

plt.figure(figsize=(16,10))

plt.subplot(2,3,1)
plt.imshow(image_rgb)
plt.title("Original")

plt.subplot(2,3,2)
plt.imshow(global_thresh, cmap='gray')
plt.title("Global")

plt.subplot(2,3,3)
plt.imshow(otsu_thresh, cmap='gray')
plt.title("Otsu")

plt.subplot(2,3,4)
plt.imshow(adaptive_thresh, cmap='gray')
plt.title("Adaptive")

plt.subplot(2,3,5)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Watershed")

plt.subplot(2,3,6)
plt.imshow(segmented_image)
plt.title("K-Means")

plt.tight_layout()
plt.show()

```

